OP-AMP CIRCUITS: Summing Circuit
Integrator Circuit
Differentiator Circuit
Current to Voltage Converter
OP-AMP as active filters

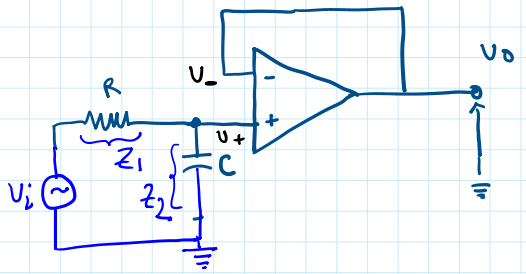
Active Filters : Low Pass Active Filter

$$V_o = V_- = V_+ = V_i * \frac{Z_2}{Z_1 + Z_2}$$

$$= V_i * \frac{(1/j\omega C)}{R + 1/j\omega C}$$

$$= V_i * \frac{1}{1 + j\omega RC}$$

$$\frac{V_o}{V_i}(\omega) = \frac{1}{1 + j\omega\tau}$$



This transfer function has characteristic of low pass filter.

$$\frac{V_o}{V_i}(\omega)\Big|_{\omega \to 0} = 1$$

$$\frac{V_o}{V_i}(\omega)\Big|_{\omega \to \infty} = 0$$

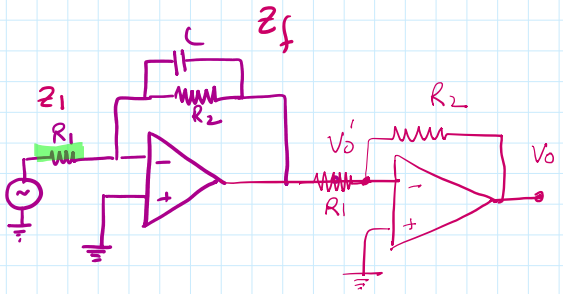
Another example of low pass filter:

$$\frac{V_o}{V_i}(\omega) = -\frac{Z_f}{Z_1}$$

$$= -\left(\frac{R_2 * \frac{1}{j\omega C}}{R_2 + \frac{1}{j\omega C}}\right) * \frac{1}{R_1}$$

$$= -\frac{(R_2/R_1)}{1 + j\omega R_2 C}$$

$$= -\frac{R_2}{R_1} * \left(-\frac{R_1}{R_2}\right) * \frac{1}{1 + j\omega R_2 C}$$

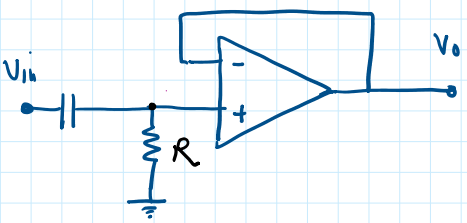


High Pass filter:

$$V_o = V_- = V_+ = \frac{R}{R + \frac{1}{j\omega C}} V_i$$

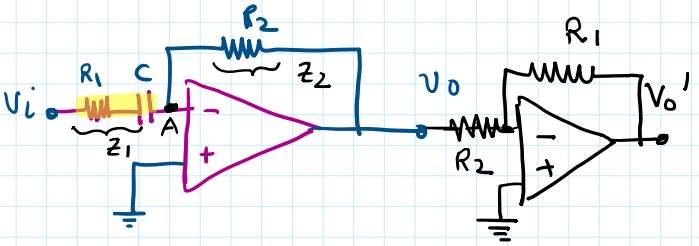
$$= \frac{j\omega RC}{1 + j\omega RC} = \frac{j\omega\tau}{1 + j\omega\tau}$$

which is transfer function of high pass filter



$$= \frac{j\omega RC}{1+j\omega RC} = \frac{j\omega\tau}{1+j\omega\tau}$$ which is transfer function of high pass filter.

Another circuit that can be used for HPF is:



$V_+ = V_-$

Using KCL at A,

$$\frac{0 - V_i}{Z_1} + \frac{0 - V_o}{Z_2} = 0$$

$$V_o = -\frac{Z_2}{Z_1} V_i = -\frac{R_2}{R_1 + \frac{1}{j\omega C}}$$

$$= -\frac{j\omega R_2 C}{1 + j\omega R_1 C}$$

$$= -\frac{R_2}{R_1} \frac{j\omega R_1 C}{1 + j\omega R_1 C}$$

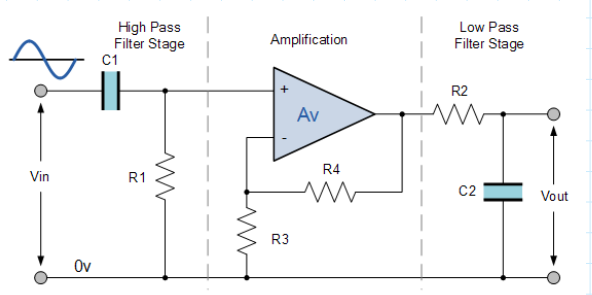
$$= -\frac{R_2}{R_1} \frac{j\omega\tau}{1+j\omega\tau}$$

Using an additional stage with $R_f = R_1$, $R = R_2$

$$\frac{V_o'}{V_{in}}(\omega) = \left(-\frac{R_2}{R_1}\right) * \left(-\frac{R_1}{R_2}\right) \frac{j\omega\tau}{1+j\omega\tau}$$

$$\boxed{\frac{V_o(\omega)}{V_{in}} = \frac{j\omega\tau}{1+j\omega\tau}}$$

Band Pass Filter



HP LP

Find the transfer function:

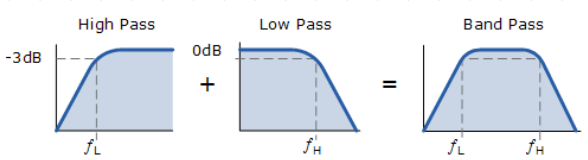
$$\frac{R_1}{R_1 + \frac{1}{j\omega C_1}} = \frac{j\omega R_1 C}{1+j\omega R_1 C} = \frac{j\omega\tau}{1+j\omega\tau}$$

$$V_- = V_+ = \left(\frac{j\omega\tau}{1+j\omega\tau}\right) V_i$$

$$\frac{V_- - 0}{R_3} + \frac{V_- - V_o}{R_4} = 0$$

$$\left(\frac{1}{R_3} + \frac{1}{R_4}\right) V_- = \frac{V_o}{R_4}$$

$$\left(\frac{R_3 + R_4}{R_3 R_4}\right) V_- = \frac{V_o}{R_4}$$



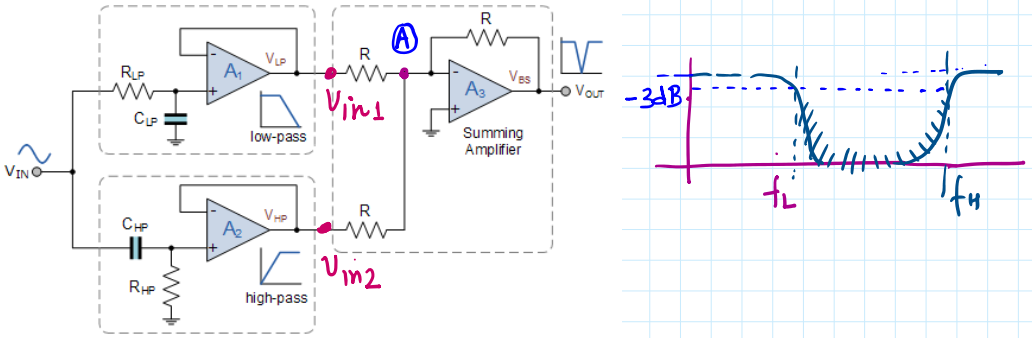
$$\left(\frac{R_3+R_4}{R_3 R_4}\right) V_- = \frac{V_o}{R_4}$$

$$V_o = \left(1+\frac{R_4}{R_3}\right) V_- = \left(1+\frac{R_4}{R_3}\right) * \frac{j\omega\tau}{1+j\omega\tau} = \frac{j\omega\tau_1}{1+j\omega\tau_1}$$

At output:

$$\frac{V_o'}{V_o} = \frac{j\omega\tau_1}{1+j\omega\tau_1} * \frac{1/j\omega C_2}{R_2+1/j\omega C_2} = \underbrace{\frac{j\omega\tau_1}{(1+j\omega\tau_1)}}_{\text{HPF}} \underbrace{\frac{1}{(1+j\omega\tau_2)}}_{\text{LPF}}$$

Band Rejection Filter



Stage 1: $V_{in1} = \frac{1}{1+j\omega\tau_L} V_{in}$

$$V_{in2} = \frac{j\omega\tau_h}{1+j\omega\tau_h} V_{in}$$

At point (A) KCL: $\frac{0-V_{in1}}{R} + \frac{0-V_{in2}}{R} + \frac{0-V_o}{R} = 0$

$$V_o = -(V_{in1} + V_{in2})$$

$$\frac{V_o}{V_{in}} = -\left(\frac{1}{1+j\omega\tau_L} + \frac{j\omega\tau_h}{1+j\omega\tau_h}\right) V_{in}$$

One more stage can be provided to nullify negative sign, i.e.

$$\frac{V_o}{V_{in}} = \left(\frac{1}{1+j\omega\tau_L} * \frac{j\omega\tau_h}{1+j\omega\tau_h}\right) V_{in}$$

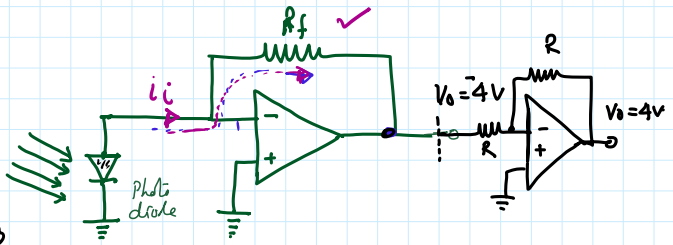
We see very interesting point, for Band Pass filtering, there is cascading of HPF and LPF

We see very interesting point, for Band Pass filtering, there is cascading of HPL and LPF.

In case of Band Rejection filter it is addition of LPF and HPF.

Some Examples

Example 1: Consider a photo diode for visible light with a sensitivity of 25 μA/mW of the incident radiation. The diode is used in the lightmeter as shown below. What is value of R_f for $V_0 = 4V$ for 50 mW of radiation.

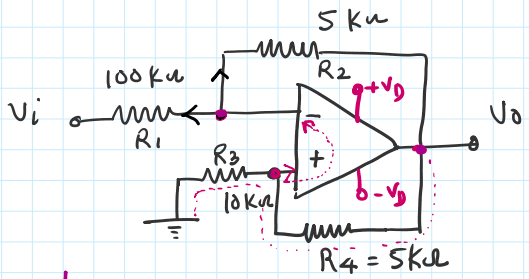


$$i_i = 25\ (\mu A/mW) * 50\ (mW)$$
$$= 1250\ \mu A$$
$$= 1.25\ mA$$

$$V_0 = -R_f i_i = -R_f * 1.25 \times 10^{-3}$$
$$V_0 = 4 = |-R_f * 1.25 \times 10^{-3}|$$
$$R_f = \frac{4}{1.25 \times 10^{-3}} = 3.2\ K\Omega$$

$$10 \times 10^{-6}\ (\mu A/mW)\ W_i * R_f = 8$$
$$10 \times 10^{-6} * W_i \times 10 \times 10^{3} = 8$$
$$W_i = \frac{8}{10 \times 10^{-6} \times 10 \times 10^{3}} = \frac{8}{10^{6} \times 10^{5}}$$
$$= \frac{8}{10^{1}} = 8 \times 10^{1} = 80\ mW$$

Q2 : Find the gain of OP-AMP Ckt.



$$\frac{V_0}{V_i} = ?$$

$$V_+ = \frac{R_3}{R_3+R_4} * V_0 = V_-$$

$$\frac{V_- - V_i}{R_1} + \frac{V_- - V_0}{R_2} = 0$$

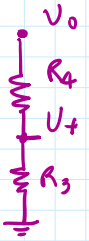
$$V_- \left[\frac{1}{R_1} + \frac{1}{R_2}\right] - \frac{V_0}{R_2} = \frac{V_i}{R_1}$$

$$V_0 \left(\frac{R_3}{R_3+R_4} * \frac{R_1+R_2}{R_1 R_2} - \frac{1}{R_2}\right) = \frac{V_i}{R_1}$$

$$V_0 \left[\frac{R_3(R_1+R_2) - R_1(R_3+R_4)}{R_1 R_2 (R_3+R_4)}\right] = \frac{V_i}{R_1}$$

$$\frac{V_0}{V_i} = \frac{R_2(R_3+R_4)}{R_2 R_3 - R_1 R_4}$$

$R_2R_3 > R_1R_4$, V_0 is +ve
$R_2R_3 < R_1R_4$, V_0 is -ve

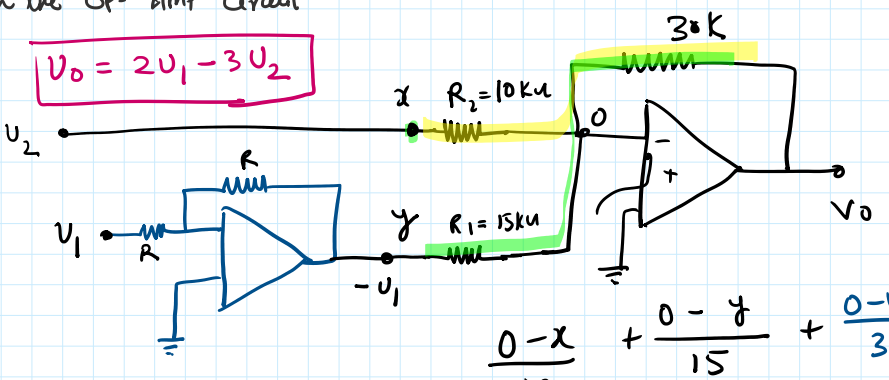


Q3 Consider a system described by

$z = 2x - 3y$ where x & y are input voltage / signals.

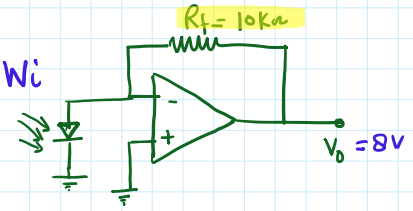
Design the OP-AMP Circuit.

$V_0 = 2V_1 - 3V_2$



$$\frac{0-x}{10} + \frac{0-y}{15} + \frac{0-V_0}{30} = 0$$

Q4 Photodiode has sensitivity of 10 μA/mW. If V_0 = 8V find the input radiation ?



$i_i = 10 \times 10^{-6}$ / mW.

Let radiation is watts be x mW.

Then $i_i = 10 \times 10^{-6} * x$ A.

$$|i_i * R_f| = 8V$$

$$10 \times 10^{-6} \times x \times 10 \times 10^{3} = 8$$

$$x = \frac{8}{10 \times 10^{-6} \times 10 \times 10^{3}}$$

$$= 8/10^{-1} = 80\text{mW}$$